PRAKTIKUM BASIS DATA

MODUL 08

Scripting SQL (Create index, Create view, Analyze Table)

Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

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Jurnal

1. a. Table department dengan primary key adalah department_id. Tabel departemen terdiri atas field-field: department_id, department_name, manager_id, dan location_id.

Query:

Worksheet | Query Builder

```
CREATE TABLE departments (
    department_id NUMBER(4) PRIMARY KEY,
    department_name VARCHAR2(30) NOT NULL,
    manager_id NUMBER(6),
    location_id NUMBER(4)
);
```

Output:

Script Output | Query Result

SQL | All Rows Fetched: 0 in 0.066 seconds

DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID

b. Table employee dengan contraint foreign key adalah department_id. Tabel employee terdiri atas field-field: employee_id, last_nama NOT NULL, email, salary, commission_pct, hire_date NOT NULL>

Query:

Worksheet | Query Builder

```
CREATE TABLE employees (
    employee_id NUMBER(6) PRIMARY KEY,
    last_name VARCHAR2(25) NOT NULL,
    email VARCHAR2(25),
    salary NUMBER(8,2),
    commission_pct NUMBER(2,2),
    hire_date DATE NOT NULL,
    department_id NUMBER(4),
    CONSTRAINT fk_dept FOREIGN KEY (department_id) REFERENCES departments(department_id)
);
```

Output:

Script Output | Query Result

SQL | All Rows Fetched: 0 in 0.067 seconds

EMPLOYEE_ID	LAST_NAME	EMAIL	SALARY	COMMISSION_PCT	HIRE_DATE	DEPARTMENT_ID

c. Tambahkan tabel-tabel lain yang saling berelasi (seperti: tabel persediaan barang, pemasok barang, dst).

Query:

Worksheet | Query Builder

```
--

CREATE TABLE locations (
    location_id NUMBER(4) PRIMARY KEY,
    street_address VARCHAR2(40),
    city VARCHAR2(30) NOT NULL,
    state_province VARCHAR2(25)
);
```

Worksheet | Query Builder

```
--

CREATE TABLE job_history (
    employee_id NUMBER(6) NOT NULL,
    start_date DATE NOT NULL,
    end_date DATE NOT NULL,
    job_id VARCHAR2(10) NOT NULL,
    department_id NUMBER(4),
    CONSTRAINT pk_job_hist PRIMARY KEY (employee_id, start_date),
    CONSTRAINT fk_emp_hist FOREIGN KEY (employee_id) REFERENCES employees(employee_id)
);
```

Output:

Script Output × | Query Result ×

SQL | All Rows Fetched: 0 in 0.061 seconds

LOCATION_ID	STREET_ADDRESS	CITY	STATE_PROVINCE

Script Output × | Query Result ×

SQL | All Rows Fetched: 0 in 0.007 seconds

EMPLOYEE_ID	START_DATE	END_DATE	JOB_ID	DEPARTMENT_ID

2. Analisis table-table dengan compute statistics

Query:

```
ANALYZE TABLE departments COMPUTE STATISTICS;
ANALYZE TABLE employees COMPUTE STATISTICS;
ANALYZE TABLE suppliers COMPUTE STATISTICS;
ANALYZE TABLE locations COMPUTE STATISTICS;
ANALYZE TABLE job_history COMPUTE STATISTICS;
```

3. Isikan data-data yang bersesuaian dengan tabel-tabel pada nomor 1 (masing-masing 10 record).

a. Locations

Query:

Worksheet | Query Builder

```
INSERT ALL
INTO locations VALUES (1000, '1297 Via Cola di Rie', 'Roma', 'IT')
INTO locations VALUES (1100, '93091 Calle della Testa', 'Venice', 'IT')
INTO locations VALUES (1200, '2017 Shinjuku-ku', 'Tokyo', 'Tokyo')
INTO locations VALUES (1300, '9450 Kamiya-cho', 'Hiroshima', 'JP')
INTO locations VALUES (1400, '2014 Jabberwocky Rd', 'Southlake', 'Texas')
INTO locations VALUES (1500, '2011 Interiors Blvd', 'South San Francisco', 'California')
INTO locations VALUES (1600, '2007 Zagora St', 'South Brunswick', 'New Jersey')
INTO locations VALUES (1700, '2004 Charade Rd', 'Seattle', 'Washington')
INTO locations VALUES (1800, '147 Spadina Ave', 'Toronto', 'Ontario')
INTO locations VALUES (1900, '6092 Boxwood St', 'Whitehorse', 'Yukon')
SELECT * FROM dual;
```

Output:

Script Output | Query Result

SQL | All Rows Fetched: 10 in 0.081 seconds

	LOCATION_ID	STREET_ADDRESS	CITY	STATE_PROVINCE
1	1000	1297 Via Cola di Rie	Roma	IT
2	1100	93091 Calle della Testa	Venice	IT
3	1200	2017 Shinjuku-ku	Tokyo	Tokyo
4	1300	9450 Kamiya-cho	Hiroshima	JP
5	1400	2014 Jabberwocky Rd	Southlake	Texas
6	1500	2011 Interiors Blvd	South S...	California
7	1600	2007 Zagora St	South B...	New Jersey
8	1700	2004 Charade Rd	Seattle	Washington
9	1800	147 Spadina Ave	Toronto	Ontario
10	1900	6092 Boxwood St	Whitehorse	Yukon

b. Departments

Query:

Worksheet | Query Builder

```
INSERT ALL
INTO departments VALUES (10, 'Administration', 200, 1700)
INTO departments VALUES (20, 'Marketing', 201, 1800)
INTO departments VALUES (30, 'Purchasing', 114, 1700)
INTO departments VALUES (40, 'Human Resources', 203, 1400)
INTO departments VALUES (50, 'Shipping', 121, 1500)
INTO departments VALUES (60, 'IT', 103, 1400)
INTO departments VALUES (70, 'Public Relations', 204, 1700)
INTO departments VALUES (80, 'Sales', 145, 1500)
INTO departments VALUES (90, 'Executive', 100, 1700)
INTO departments VALUES (100, 'Finance', 108, 1700)
SELECT * FROM dual;
```

Output:

Script Output | Query Result

SQL | All Rows Fetched: 10 in 0.007 seconds

	DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
1	10	Administration	200	1700
2	20	Marketing	201	1800
3	30	Purchasing	114	1700
4	40	Human Resources	203	1400
5	50	Shipping	121	1500
6	60	IT	103	1400
7	70	Public Relations	204	1700
8	80	Sales	145	1500
9	90	Executive	100	1700
10	100	Finance	108	1700

c. Employees

Query:

Worksheet | Query Builder

```
INSERT ALL
INTO employees VALUES (101, 'Budiman', 'BUDIMAN', 17000, NULL, TO_DATE('2021-09-21','YYYY-MM-DD'), 90)
INTO employees VALUES (102, 'Suryono', 'SURYONO', 17000, NULL, TO_DATE('2022-01-13','YYYY-MM-DD'), 90)
INTO employees VALUES (103, 'Prabowo', 'PRABOWO', 9000, NULL, TO_DATE('2022-01-03','YYYY-MM-DD'), 60)
INTO employees VALUES (104, 'Lestari', 'LESTARI', 6000, NULL, TO_DATE('2023-05-21','YYYY-MM-DD'), 60)
INTO employees VALUES (107, 'Santoso', 'SANTOSO', 4200, NULL, TO_DATE('2024-02-07','YYYY-MM-DD'), 60)
INTO employees VALUES (145, 'Wijaya', 'WIJAYA', 14000, 0.4, TO_DATE('2021-10-01','YYYY-MM-DD'), 80)
INTO employees VALUES (146, 'Saputra', 'SAPUTRA', 13500, 0.3, TO_DATE('2021-01-05','YYYY-MM-DD'), 80)
INTO employees VALUES (147, 'Hidayat', 'HIDAYAT', 12000, 0.3, TO_DATE('2022-03-10','YYYY-MM-DD'), 80)
INTO employees VALUES (148, 'Kusuma', 'KUSUMA', 11000, 0.3, TO_DATE('2023-10-15','YYYY-MM-DD'), 80)
INTO employees VALUES (149, 'Anggraini', 'ANGGRAINI', 10500, 0.2, TO_DATE('2024-01-29','YYYY-MM-DD'), 80)
SELECT * FROM dual;
```

Output:

Query Result

All Rows Fetched: 10 in 0.008 seconds

	EMPLOYEE_ID	LAST_NAME	EMAIL	SALARY	COMMISSION_PCT	HIRE_DATE	DEPARTMENT_ID
1	101	Kochhar	NKOCHHAR	17000	(null)	21-09-2021	90
2	102	De Haan	LDEHAAN	17000	(null)	13-01-2022	90
3	103	Hunold	AHUNOLD	9000	(null)	03-01-2022	60
4	104	Ernst	AERNST	6000	(null)	21-05-2023	60
5	107	Lorentz	DLORENTZ	4200	(null)	07-02-2024	60
6	145	Russell	JRUSSELL	14000	0,4	01-10-2021	80
7	146	Partners	KPARTNER	13500	0,3	05-01-2021	80
8	147	Errazuriz	AERRAZUR	12000	0,3	10-03-2022	80
9	148	John	GCAMBRAU	11000	0,3	15-10-2023	80
10	149	Peter	ELZLOTKE	10500	0,2	29-01-2024	80

d. Job_History

Query:

Worksheet | Query Builder

```
INSERT ALL
INTO job_history VALUES (101, TO_DATE('2015-09-21','YYYY-MM-DD'), TO_DATE('2016-10-27','YYYY-MM-DD'), 'AC_ACCOUNT', 110)
INTO job_history VALUES (101, TO_DATE('2016-10-28','YYYY-MM-DD'), TO_DATE('2019-03-15','YYYY-MM-DD'), 'AC_MGR', 110)
INTO job_history VALUES (102, TO_DATE('2011-01-13','YYYY-MM-DD'), TO_DATE('2016-07-24','YYYY-MM-DD'), 'IT_PROG', 60)
INTO job_history VALUES (103, TO_DATE('2020-01-01','YYYY-MM-DD'), TO_DATE('2021-12-31','YYYY-MM-DD'), 'IT_PROG', 60)
INTO job_history VALUES (104, TO_DATE('2021-01-01','YYYY-MM-DD'), TO_DATE('2022-12-31','YYYY-MM-DD'), 'ST_CLERK', 50)
INTO job_history VALUES (107, TO_DATE('2022-01-01','YYYY-MM-DD'), TO_DATE('2023-12-31','YYYY-MM-DD'), 'SA_REP', 80)
INTO job_history VALUES (145, TO_DATE('2018-01-01','YYYY-MM-DD'), TO_DATE('2019-12-31','YYYY-MM-DD'), 'SA_MAN', 80)
INTO job_history VALUES (146, TO_DATE('2019-01-01','YYYY-MM-DD'), TO_DATE('2020-12-31','YYYY-MM-DD'), 'SA_REP', 80)
INTO job_history VALUES (147, TO_DATE('2020-01-01','YYYY-MM-DD'), TO_DATE('2021-12-31','YYYY-MM-DD'), 'SA_REP', 80)
INTO job_history VALUES (148, TO_DATE('2021-01-01','YYYY-MM-DD'), TO_DATE('2022-12-31','YYYY-MM-DD'), 'SA_REP', 80)
SELECT * FROM dual;
```

Output:

Query Result

All Rows Fetched: 10 in 0.008 seconds

	EMPLOYEE_ID	START_DATE	END_DATE	JOB_ID	DEPARTMENT_ID
1	101	21-09-2015	27-10-2016	AC_ACCOUNT	110
2	101	28-10-2016	15-03-2019	AC_MGR	110
3	102	13-01-2011	24-07-2016	IT_PROG	60
4	103	01-01-2020	31-12-2021	IT_PROG	60
5	104	01-01-2021	31-12-2022	ST_CLERK	50
6	107	01-01-2022	31-12-2023	SA_REP	80
7	145	01-01-2018	31-12-2019	SA_MAN	80
8	146	01-01-2019	31-12-2020	SA_REP	80
9	147	01-01-2020	31-12-2021	SA_REP	80
10	148	01-01-2021	31-12-2022	SA_REP	80

e. Suppliers

Query:

Worksheet | Query Builder

```
INSERT ALL
INTO suppliers VALUES (1, 'PT. Maju Mundur', 'Budi', 'Jakarta')
INTO suppliers VALUES (2, 'CV. Sumber Makmur', 'Siti', 'Bandung')
INTO suppliers VALUES (3, 'PT. Teknologi Jaya', 'Agus', 'Surabaya')
INTO suppliers VALUES (4, 'Indo Logistic', 'Dewi', 'Semarang')
INTO suppliers VALUES (5, 'Surya Kencana', 'Eko', 'Yogyakarta')
INTO suppliers VALUES (6, 'Bakti Nusa', 'Hendra', 'Medan')
INTO suppliers VALUES (7, 'Cahaya Abadi', 'Rina', 'Makassar')
INTO suppliers VALUES (8, 'Mega Industri', 'Andi', 'Balikpapan')
INTO suppliers VALUES (9, 'Sinar Baru', 'Wati', 'Denpasar')
INTO suppliers VALUES (10, 'Prima Sarana', 'Bambang', 'Solo')
SELECT * FROM dual;
```

Output:

Query Result

All Rows Fetched: 10 in 0.072 seconds

	SUPPLIER_ID	SUPPLIER_NAME	CONTACT_NAME	CITY
1	1	Logitech	Mark	Lausanne
2	2	Microsoft	Bill	Redmond
3	3	Oracle Corp	Larry	Redwood City
4	4	Intel	Pat	Santa Clara
5	5	Samsung	Lee	Seoul
6	6	Sony	Ken	Tokyo
7	7	Dell	Michael	Round Rock
8	8	HP	Enrique	Palo Alto
9	9	Cisco	Chuck	San Jose
10	10	Lenovo	Yang	Beijing

4. Analisis table-table dengan compute statistics

Query:

```
ANALYZE TABLE departments COMPUTE STATISTICS;
ANALYZE TABLE employees COMPUTE STATISTICS;
ANALYZE TABLE suppliers COMPUTE STATISTICS;
ANALYZE TABLE locations COMPUTE STATISTICS;
ANALYZE TABLE job_history COMPUTE STATISTICS;
```

5. Buat view empvu80 yang berisi id_number, name, salary, department_id dari pegawai yang bekerja di department = 80.

Query:

```
CREATE OR REPLACE VIEW empvu80 AS
SELECT employee_id AS id_number, last_name AS name, salary, department_id
FROM employees
WHERE department_id = 80;
```

6. Kemudian tampilkan struktur dari view empvu80

Output:

Script Output | Query Result

SQL | All Rows Fetched: 5 in 0.009 seconds

	ID_NUMBER	NAME	SALARY	DEPARTMENT_ID
1	145	Russell	14000	80
2	146	Partners	13500	80
3	147	Errazuriz	12000	80
4	148	John	11000	80
5	149	Peter	10500	80

7. Buat non-unique index pada kolom FOREIGN KEY yang ada pada tabel EMPLOYEE

Query:

```
CREATE INDEX emp_dept_idx ON employees(department_id);
```

Output:

```
Name          Null?    Type
------------- -------- ------------
ID_NUMBER     NOT NULL NUMBER(6)
NAME          NOT NULL VARCHAR2(25)
SALARY                 NUMBER(8,2)
DEPARTMENT_ID          NUMBER(4)
>>Query Run In:Query Result

Index EMP_DEPT_IDX created.
```